D 70326 (Pages : 3) Name.....

Reg. No.....

FIFTH SEMESTER B.Sc. DEGREE EXAMINATION, NOVEMBER 2019

(CUCBCSS—UG)

Mathematics

MAT 5D 19—MATHEMATICS FOR SOCIAL SCIENCES

Time : Two Hours Maximum : 40 Marks

Section A

Answer all the **six** *questions.*
Each question carries **1** *mark.*

1. Solve $\frac{x}{3} - 16 = \frac{x}{12} + 14$.

2. Find the slope m of the linear function passing through $(-1, 15)$ and $(3, 6)$.

3. Find $\lim_{x \to 2} \sqrt{6x^3 + 1}$.

4. Define concavity and convexity.

5. Find the marginal cost if the total cost function $TC = Q^3 + 7Q + 23$.

6. Test whether the function $f(x) = 3x^3 - 14x + 5$ is increasing or decreasing at $x = 4$.

(6 × 1 = 6 marks)

Section B

Answer any **five** *out of seven questions.*
Each question carries **2** *marks.*

7. Solve the quadratic equation $3x^2 - 41x + 26 = 0$.

8. Which of the following equations are functions and why ?

   (a) $y = -2x + 7$.

   (b) $y^2 = x$.

9. Verify whether the function $f(x) = \frac{x^2 + 3x + 12}{x - 3}$ is continuous at $x = 4$.

10. Find the derivative of $y = \frac{4x^3}{1 - 3x}$.

11. If $\log_2 x + \log_4 x + \log_{16} = \frac{21}{4}$, find x.

12. Evaluate $\int_1^{e^2} \left(\frac{dx}{x(1 + \log_e x)} \right) dx$.

13. A firm's marginal revenue function is $MR = 20e^{-x/10}\left(1 - \frac{x}{10}\right)$. Find the corresponding demand function.

(5 × 2 = 10 marks)

Section C

Answer any **three** *out of five questions.*
Each question carries **4** *marks.*

14. If $y = (\log x)^{\log x}$, find $\frac{dy}{dx}$.

15. Evaluate $\lim_{x \to \infty} \sqrt{\frac{x^2 - 5x}{x^3 + x - 2}}$.

16. Find the cross partial derivatives z_{xy} and z_{yx} for the function $z = x^{0.3}\, y^{0.5}$.

17. Find the relative extrema for the function $f(x) = -7x^2 + 126x - 23$.

18. Find the marginal revenue function and evaluate them at Q = 4 and Q = 10 if the demand function is Q = 36 – 2P.

(3 × 4 = 12 marks)

Section D

Answer any **two** *out of three questions.*
Each question carries 6 *marks.*

19. Maximize profit π for a firm, given total revenue $R = 4000\,Q - 33\,Q^2$ and total cost $C = 2Q^3 - 3Q^2 + 400Q + 5000$.

20. Use integration by parts to integral $\int_1^e x \log_x dx$.

21. Suppose that a firms total cost is given by the function $C = f(q) = 8 + 2q + 0.25q^2$, where C denotes the total cost in dollars and q denotes the total quantity of the good produced. Find the marginal cost of the firm when $q = 10$, and the percentage rate of change of total cost with respect to q when $q = 10$ and interpret the result.

(2 × 6 = 12 marks)